

LYA CHIÑAS (SHE/HER)

PHD STUDENT | MACHINE LEARNING FOR BIOLOGY

📍 Toronto, Canada

✉ lya.chinas@mail.utoronto.ca

🌐 [Lya Chiñas Serrano](#)

EDUCATION

Ph.D. STUDENT

University of Toronto,
Dept. of Chemical
Engineering

Toronto, Canada • 2024
– Pres.

B.S. IN BIOTECHNOLOGY ENGINEERING

Tecnológico de
Monterrey

Puebla, Mexico • 2020 –
2024

GPA 4.0, First in Class
Honors

SKILLS

Languages: Python,
MATLAB, HTML, CSS

Tools: PyTorch,
TensorFlow, scikit-learn,
NumPy, Pandas,
Matplotlib, Git,
Bash/Shell, Slurm,
LaTeX

Content Creation:
Adobe (Photoshop,
Illustrator, Premiere Pro),
Affinity (Photo, Designer,
Publisher)

LANGUAGES

Spanish • Native

English • Advanced •
TOEFL C1

French • Elementary •
DELTA A2

FELLOWSHIPS

**Massey College Junior
Fellow**

2024 – Present

**International Society
for Computational
Biology**

2026 – Present

EXPERIENCE

GRADUATE STUDENT | LABORATORY FOR METABOLIC SYSTEMS ENGINEERING

University of Toronto • Sept 2024 – Pres.

- Creating and evaluating ML/DL models for the prediction of enzyme parameters and *E. coli* phenotypes.
- Building end-to-end pipelines, with multi-omics data compilation, model training, and evaluation.

TEACHING ASSISTANT | UNIVERSITY OF TORONTO

CHE1148 'Artificial Intelligence for Applied Chemistry and Chemical Engineering' • Jan 2025 – Pres.

CHE1147 'Chemical Data Science and Engineering' • Sep 2025 – Dec 2025

CHE222 'Process Dynamics: Modeling, Analysis and Simulation' • Jan 2025 – May 2025

- Conducted tutorials and practical laboratories on data science, machine learning, and numerical methods for chemical engineering.

GLOBALINK MENTOR | MITACS

Mitacs Inc. • Toronto, Canada • Apr 2026 – Present • Apr 2025 – Oct 2025

- Provided on-site mentorship to international undergraduate students during their 12-week stay in Canada with the Globalink Research Internship (GRI) program.

RESEARCH INTERN | LABORATORY FOR METABOLIC SYSTEMS ENGINEERING

University of Toronto • Toronto, Canada • Jul 2023 – Oct 2023

- Conducted a computational research internship sponsored by the Mitacs GRI program.
- Developed a Python framework for the reconstruction of Genome-Scale Metabolic Models.

ORGANIZING COMMITTEE | LATIN FOOD 2022

Mexican Association of Food Science (AMECA, A.C.) • Puebla, Mexico • Jun 2022 – Nov 2022

- Participated in the organization of one of Mexico's largest biotechnology congresses, leading the review of +500 abstracts of the presented research.

PUBLICATIONS

R. A. Barghout, M. Kirby, A. Zheng, L. Chinas, M. Mohammadi, Z. Xu, B. Sanchez-Lengeling, and R. Mahadevan, "Machine Learning for Sequence-to-Function Approaches," in *Machine Learning and Big Data-enabled Biotechnology*, John Wiley & Sons, 2026, pp. 103-129.

Barghout, R. A., Serrano, L. C., Sanchez-Lengeling, B., & Mahadevan, R. (2026). kinGEMs: A Robust and Scalable Framework for Resource-Constraint Models through Stochastic Tuning of Deep Learning-Predicted Kinetic Parameters. *bioRxiv (Cold Spring Harbor Laboratory)*. <https://doi.org/10.64898/2026.03.14.711833>

PRESENTATIONS

CONFERENCE ON CONSTRAINT-BASED RECONSTRUCTION AND ANALYSIS

Potsdam, GER • 2026 • Poster presentation

San Diego, USA • 2024 • Oral & poster presentations

AMERICAN INSTITUTE OF CHEMICAL ENGINEERS ANNUAL MEETING

San Diego, USA • 2024 • Oral presentation

INTERNATIONAL CONFERENCE ON LEARNING REPRESENTATIONS (ICLR) 2025

Singapore • 2025 • Learning Meaningful Representations of Life Workshop

AWARDS

CONNAUGHT INTERNATIONAL SCHOLARSHIP FOR DOCTORAL STUDENTS

University of Toronto • 2024 – 2029

SECIHTI SCHOLARSHIP FOR GRADUATE STUDIES

Mexican Secretary of Science, Humanities, and Innovation • 2024 – 2029